

COSC2148/COSC3130 — Computing Research & Project Preparation

Assignment #1: Team Formation, Project Selection, and List of Papers

Semester 1, 2026

Team Members

Student ID	Name	Role
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Selected Topic

Using Physiological Cues to Improve Empathy and Human-AI Interaction in a Mixed Reality Environment

Topic Description

The goal of this project is to investigate how physiological signals such as eye gaze, facial expressions, GSR, and heart rate can be used as input and output to support empathy during human-AI interactions in a Mixed Reality environment. We will use VR headsets and GSR sensors as hardware combined with AI/ML to understand how an AI agent can influence human behaviours in the XR world.

List of Relevant Research Papers

- [1] Gebhardt, S., Hensen, B., & Klompmaker, F. (2024). *Perceived Empathy in Mixed Reality: Assessing the Impact of Empathic Agents' Awareness of User Physiological States*. In *Proceedings of the 2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE. <https://ieeexplore.ieee.org/document/10765390>
- [2] Abadi, M. K., Subramanian, R., Kia, S. M., Avesani, P., Patras, I., & Sebe, N. (2018). *Affective computing in virtual reality: emotion recognition from brain and*

heartbeat dynamics using wearable sensors. Scientific Reports, 8(1), 13843. <https://doi.org/10.1038/s41598-018-32063-4>

- [3] Kragting, M., Rieffe, C., Cruciani, F., & Laban, G. (2025). *Empathetic Conversational Agents: Utilizing Neural and Physiological Signals for Enhanced Empathetic Interactions. International Journal of Human-Computer Interaction*. <https://doi.org/10.1080/10447318.2025.2540500>
- [4] Roo, J. S., Gervais, R., & Hachet, M. (2025). *Empathic Extended Reality in the Era of Generative AI. Extended Reality (XR) and Affective Computing*. <https://www.sciexplor.com/articles/ec.2025.0009>
- [5] Piumsomboon, T., Lee, G. A., Irlitti, A., Ens, B., Thomas, B. H., & Billinghamurst, M. (2020). *AffectivelyVR: Towards VR Personalized Emotion Recognition*. In *Proceedings of the 26th ACM Symposium on Virtual Reality Software and Technology (VRST '20)*. ACM. <https://doi.org/10.1145/3385956.3422122>

AI Attribution Declaration

This work was primarily human-created. AI (Claude by Anthropic) was used to support brainstorming of research topics and to identify relevant academic papers. All AI-generated suggestions were reviewed, verified, and approved by the team. The following model(s) or application(s) were used: Claude (Anthropic).

AIA Partial AI Use v1.0

Appendix: Documentation of Generative AI Use

GenAI Tool Used	Claude (Anthropic, claude.ai)
Purpose	Brainstorming research topic alignment; identifying and verifying relevant academic papers.
Prompts Given	(1) “The topic we are going to work on will be HCI – using AI for sports training.” (2) “Changing topic to Using Physiological Cues to Improve Empathy and Human-AI Interaction in a Mixed Reality Environment.” (3) Provided the official project description for paper alignment.
Number of Prompts	3 key prompts
Summary of AI Output	Claude suggested 5 relevant research papers with full references, mapped to the key concepts in the project description (physiological signals, empathy, MR/VR, AI agents, HCI).
Verification	Each paper was checked against its DOI/URL to confirm it exists and is relevant to the topic. Author names and publication details were verified.
Usage / Modification	The project description was written by the team based on the official topic list. Paper suggestions were reviewed and selected by the team leader.
Reflection	The tool was effective for quickly surfacing relevant literature. However, AI-generated author names required manual verification, as some were incomplete or approximate.